

AKASH AGARWAL

+1-623-281-6287 | akashagarwal2908@gmail.com | [linkedin.com/in/akash-agarwal2908](https://www.linkedin.com/in/akash-agarwal2908) | github.com/Akash-Agarwal-X

PROFESSIONAL EXPERIENCE

Sharedmemory.ai | Co-Founder & Founding Engineer

Nov 2024 – Present · Remote

- Architected a **distributed, multi-tiered SaaS platform** for AI memory persistence — designed for high availability and horizontal scalability across independent agent runtimes and IDE integrations.
- Designed the core orchestration engine with **concurrent multi-threaded coordination**, conflict resolution via vector-clock ordering, and trust-tiered durability guarantees for long-term context retention.
- Built fault-tolerant backend services using **OOP design patterns** (strategy, observer, chain-of-responsibility) ensuring reliable consistency across distributed nodes under partial failure conditions.

Amazon Web Services (AWS) | SDE Intern

May 2025 – Aug 2025 · Seattle, WA

- Designed a **high-throughput, low-latency distributed storage system** on DynamoDB with encryption-aware compression, achieving **97% storage efficiency** and cutting monthly costs from **\$1,089 to \$24.75**.
- Engineered epoch-based indexing for **O(1) writes and sub-millisecond point queries** across billions of ELB state change records, maintaining durability and consistency across distributed AWS regions.
- Built **fault-tolerant recovery subsystems** with automated replay and consistency verification, ensuring zero data loss and high availability across all tested failure and partition scenarios.
- Overhauled ELB Radar compliance auditing tooling, reducing engineer investigation time from **4+ hours to under 2 minutes** and improving operational observability across the fleet.

Arizona State University | Software Engineer, Research Systems

Mar 2025 – May 2026 · Tempe, AZ

- Built and deployed **Containerized, Kubernetes-orchestrated** backend services on GCP with automated health checks and rolling restarts, enabling zero-downtime upgrades across 3 university systems.
- Designed REST lifecycle management APIs and CAS/SAML SSO for a self-service portal serving **50+ research teams**, eliminating manual provisioning and reducing turnaround from days to minutes.

Neova Solutions | Software Engineer

Apr 2023 – Aug 2024 · Pune, India

- Designed a Python monitoring framework applying **SOLID principles and OOP design patterns** (factory, decorator, observer) to achieve **27% higher defect detection** and **96.6% on-time release rate**.
- Mentored 2 engineers on scalable system design and reliability practices; drove team-wide adoption of automated testing, reducing regression cycle time by over 60%.

Tata Consultancy Services (TCS) | Software Engineer

Jun 2021 – Aug 2022 · Pune, India

- Architected **multi-tiered Java/Spring Boot microservices** for high-throughput transaction processing; reduced p95 latency by **36%** through multi-threaded concurrency tuning, query optimization, and connection pooling.
- Led **zero-downtime migration of 15+ legacy monoliths** to containerized microservices on AWS EKS, enforcing blue-green deployments and cutting annual infrastructure spend by **40%**.

EDUCATION

Arizona State University — M.S. in Software Engineering

Tempe, AZ · GPA: 4.0/4.0 · May 2026

University of Pune — M.S. & B.S. in Computer Science

Pune, India · May 2023 & May 2021

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C++, C, SQL

Backend & Distributed Systems: Java Spring Boot, Microservices, REST APIs, GraphQL, WebSocket, Apache Kafka, Apache Flink, Event-Driven Architecture, High Availability, Fault Tolerance, Concurrency, Multi-threading, OOP Design Patterns, Scalable Systems

Cloud & Infrastructure: AWS (EKS, EC2, DynamoDB, Lambda, S3, RDS, CDK, CloudWatch, IAM, Route53), GCP, Kubernetes, Docker, Terraform, Linux, CI/CD, Blue-Green Deployments

Data, Observability & Tools: PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, Schema Design, Caching, Indexing, Distributed Tracing, JUnit, PyTest, JMeter, Git, React, Node.js, Datadog, Cursor, Claude Code, GitHub Copilot

PROJECTS

RAG, Claude, Vector Search, GCP | RxRefactor

April 2026

- Built an AI-powered insurance coverage analysis system using RAG with Claude over 6,100+ policy chunks from 5 major US payers, reducing drug coverage lookup time from hours to seconds at \$0.01/query.

Python, Multi-Agent Systems, Shared Memory | MACE (Multi-Agent Coordination Engine)

Jan 2026 – May 2026

- Developed a multi-agent orchestration engine where a central coordinator decomposes tasks, routes execution across agents, and resolves conflicts including duplicate work, dependency violations, and resource contention using shared memory coordination.